How did Driver and Navigator roles improve code quality?

The Driver focused on writing the code properly and making sure the syntax was correct. At the same time, the Navigator kept checking the logic and thinking about possible mistakes or edge cases. Because of this, we were able to catch errors immediately instead of finding them later. The Navigator also suggested small improvements, like checking if a course already exists or handling invalid input, which made the program more reliable.

Was role switching beneficial? Why?

Yes, role switching was very helpful. When we switched roles, both of us got a chance to understand the code fully. It also made the activity fair and more engaging. As the new Driver, it helped in understanding the logic deeply, and as the new Navigator, it was easier to identify improvements with a fresh perspective.

How did Pair Programming reduce errors?

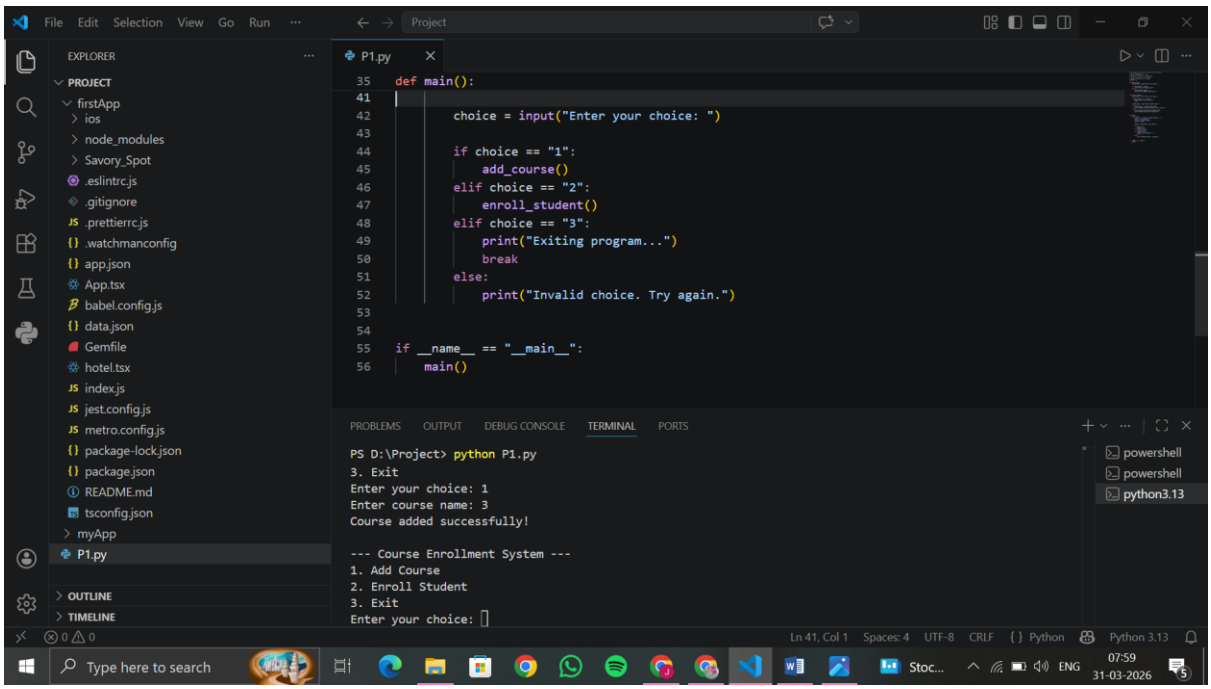
Since one person was always reviewing while the other was coding, mistakes were caught immediately. Logical errors, missing conditions, and small bugs were identified during coding itself. This saved time and reduced debugging later.

How does Pair Programming support Agile values?

Pair Programming supports Agile values because it encourages teamwork, communication, and shared responsibility. It focuses on collaboration instead of working alone. It also promotes continuous feedback, which improves overall quality.

Reflection

This lab helped me understand how Pair Programming actually works in a practical way. At first, it felt a little unusual to explain every line while coding, but it slowly improved our communication and clarity. As the Driver, I focused on writing correct code, while the Navigator continuously reviewed the logic and suggested improvements. This helped us avoid mistakes and think more clearly about edge cases. Role switching was especially useful because both of us got equal involvement in the project. It ensured that we both understood the full structure of the program instead of just one part. We were able to detect errors early and improve the code step by step. Thus this activity showed that collaboration leads to better results. It clearly demonstrated Agile principles like teamwork, communication, and continuous feedback. It was a productive and meaningful learning experience.

Role Switching Evidence

Initially, Daksh acted as the Driver and implemented the dictionary structure, Add Course function, and Enroll Student function. After Iteration 1, roles were switched. Janhavi became the Driver and implemented the Drop Course functionality and View Enrolled Courses feature. The Navigator reviewed naming conventions, indentation, and suggested minor improvements in error handling. Both members actively participated in discussion and respected the Driver–Navigator discipline during the session.